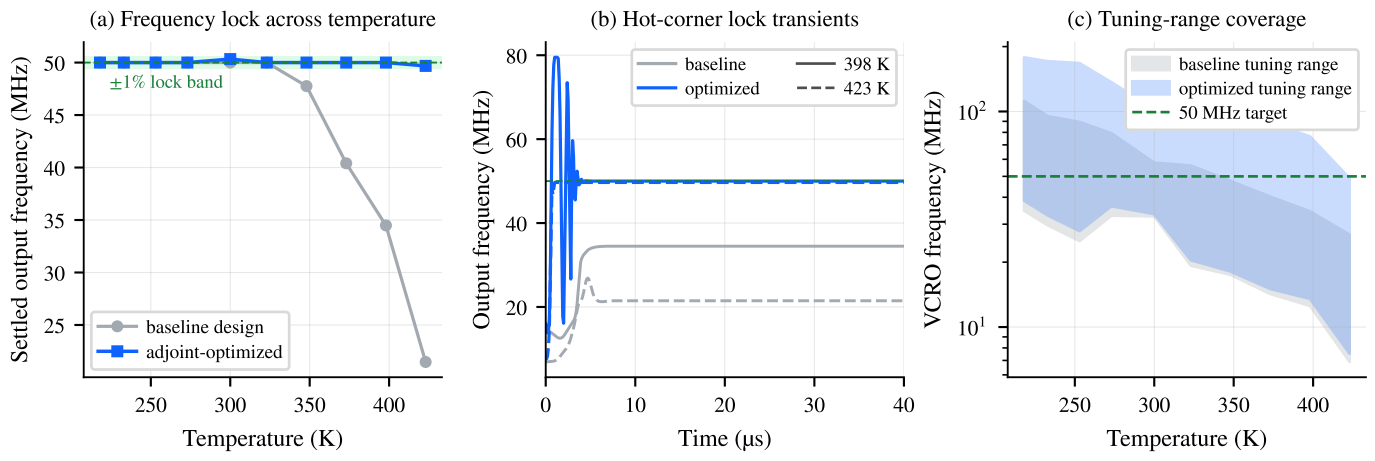
(a) Frequency lock across temperature
Settled output frequency (MHz)
50
45
40
35
30
25
±1% lock band
baseline design
adjoint-optimized
250
300
350
400
Temperature (K)
(b) Hot-corner lock transients
Output frequency (MHz)
80
60
40
20
baseline
optimized
398 K
423 K
0
10
20
30
40
Time (µs)
(c) Tuning-range coverage
VCRO frequency (MHz)
10^2
10^1
baseline tuning range
optimized tuning range
50 MHz target
250
300
350
400
Temperature (K)